Diwesh Sharma

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Summary

Proficient in machine learning, deep learning, and NLP, building end-to-end pipelines using Python, PyTorch, Scikit-learn, XGBoost, and Streamlit. Experienced in data preprocessing, feature engineering, model evaluation, and deployment via FastAPI and Docker. Built and deployed projects achieving 94% accuracy in NLP classification, 0.97 ROC-AUC in anomaly detection, and 93%+ accuracy in CNN-based image classification. Strong foundation in Data Structures, Statistics, and Algorithms. Actively seeking AI/ML Engineer roles.

Education

Dronacharya College of Engineering — Gurugram, Haryana
Bachelor of Technology, Computer Science (AI & ML) — CGPA: 8.0/10.0 — *May 2023 – May 2027*

A.V.R Public School — Delhi
Senior Secondary (XII), Science, CBSE — 89% — *2023*

A.V.R Public School — Delhi
Secondary School (X), CBSE — 93% — *2021*

Skills

Programming Languages: Python, SQL, C++

ML/DL Frameworks: PyTorch, Scikit-learn, XGBoost, CNNs (Transfer Learning), LSTMs/RNNs, Transformers

Libraries & Tools: NumPy, Pandas, Matplotlib, Seaborn, NLTK, spaCy, KeyBERT, LangChain

Data Science: NLP, Feature Engineering, Statistical Analysis, Model Evaluation, Data Visualization

Technologies: FastAPI, Streamlit, Docker, Git, GitHub, Jupyter Notebook, CUDA/GPU Training

Projects

Student Placement Prediction System | *Python, Scikit-learn, XGBoost, FastAPI, Groq API, Render* — Demo Link

- Trained and benchmarked 6 models (Logistic Regression, Random Forest, XGBoost for classification; Linear Regression, Random Forest, XGBoost Regressor for salary prediction) on 100K student records
- XGBoost selected as best classifier (Recall: 0.63) and Linear Regression as best regressor (R2: 0.79); deployed cascaded pipeline with shared preprocessor to eliminate data leakage
- Integrated Groq LLM (Llama 3.1) for personalized weak-area suggestions; deployed full-stack on Render with FastAPI backend and Streamlit frontend

Credit Card Fraud Detection | *Python, XGBoost, SMOTE, Scikit-learn, Imbalanced-learn* — GitHub

- Achieved 98% recall and 0.97 ROC-AUC on highly imbalanced dataset (0.17% fraud rate) using XGBoost with hyperparameter tuning
- Applied SMOTE oversampling to address class imbalance, improving fraud detection rate by 35%
- Benchmarked Logistic Regression, Decision Tree, and XGBoost; optimized for recall and F1-score to minimize false negatives

Quillify – AI Writing Assistant | *Gemini API, LangChain, KeyBERT, spaCy, FastAPI, Streamlit* — In Progress

- Architecting a multi-module AI writing assistant integrating Gemini API, LangChain, and spaCy across four tools: Blog SEO Analyzer, Paragraph Enhancer, Resume Optimizer, and Tone Rewriter
- Designing keyword-extraction and content-scoring logic with KeyBERT and spaCy, backed by a planned FastAPI + Streamlit deployment architecture

Heart Disease Prediction | *Python, Scikit-learn, Streamlit, Plotly* — Demo Link

- Deployed Streamlit app predicting heart disease risk on UCI dataset (303 patients, 13 features) achieving 84.49% accuracy with interactive Plotly dashboard

Certifications

Certified Generative AI Professional – Oracle

Explore Machine Learning using R – Infosys